

Written Analysis (as required: roughly half a page)

The 700 MHz link reaches 1358 km versus 34 km for 28 GHz, a 40x range ratio that matches the frequency ratio (28000/700), since path loss depends on frequency only through $20 \cdot \log_{10}(f)$.

Task 1: raising 28 GHz transmit gain to 24 dBi adds 9 dB, recovering range by $10^{(9/20)} = 2.82x$, to 96 km.

Task 2: adding 3.5 GHz gives a 272 km range, between the other two bands, in proportion to its frequency.

Task 3: an 8 dB shadowing margin shrinks every band's range to 40% of its original value, since the same dB penalty produces the same distance ratio at any frequency.

Task 4: despite P_{RX} falling 20 dB per distance decade, Shannon capacity falls only slowly because of the logarithm in $C = B \cdot \log_2(1+SNR)$: 700 MHz capacity drops from 288 to 155 Mbps between 0.1 and 10 km.

Task 5: doubling bandwidth raises capacity far more ($\times 1.92$) than doubling power ($\times 1.04$), because bandwidth scales capacity linearly while power enters only inside the logarithm.

Task 6: TDD's duty cycle is roughly $(DL \text{ slots} \div \text{total slots}) \times (1 - \text{guard fraction})$, so its instantaneous link budget matches FDD's but its average throughput is lower, since it only transmits during its own slots.